

BoxNet: Fine-grained Action Classification in Boxing

Qirui Yang

School of Automation

Southeast University

Nanjing, China

stevenyang@seu.edu.cn

Siyu Xia

School of Automation

Southeast University

Nanjing, China

xsy@seu.edu.cn

Abstract—The skeleton data is widely used in general action recognition since it is robustly distinguish the action operator and complex background. When it comes to more fine-grained scenario like those in sports competitions, such characteristic of skeleton data could further enlarge differences of actions among varied classes and reduce negative effects from background to achieve better recognition result. However, the problem of coarse video record which severely deteriorate the validity of neural network is still to be solved. In this work, we present an improved graph convolutional network(GCN) methods of fine-grained technical action classification based on previous works, where early fusion and parameterized adjacency matrix are also adopted. An interpolation method is firstly applied in action classification to elevate the performance of networks with coarse input. Moreover, a new dataset about boxing established for fine-grained action classification task. Our model is verified on NTU-RGB+D and tested on established boxing dataset, and exceeds common GCN methods.

Index Terms—Graph convolution, fine-grained action classification, skeleton dataset

I. INTRODUCTION

Advanced sports training task requires precised and general analytical tools as most human coaches can hardly provide highly unanimous and quantify feedback of technical action [1], [2]. The first step of sports analytic is the classification of technical action. Action classification in such specialized scenarios is distinguished from that in general scene as the inter-class difference usually be smaller than inner-class difference, which means the application in specialized fields should be considered as a kind of fine-grained classification. Take technical action from boxing as example, the differences of jolt and swing merely from the level of body inclination and the direction of palm, not in the angle and position of limbs, which is more obvious in general human action. For example, During jolt body inclines to the front and palm is toward to the downside while body inclines to the left or right and palm is toward to the left side or the right side during swing. Conventional action classification methods designed and trained on general dataset can hardly satisfied the need of this kind of fine-grained classification in specialized fields, as they concentrate on every node of human body which makes the irrelevant part influence the final estimation of fine-grained action.

Another primary feature of sports technical action is its high velocity, which makes it difficult to capture all distinguished key-frames by cameras with recording capability of 60fps or lower. For example, the locus of fist and elbow during boxing can not be precisely detected by common camera as the speed should be high and the long exposure duration may make the image obscure. Available dataset may be lack of some key-frames of an action, as most of them only have frame rate of less than 30fps, which could influence the accuracy of key-frame's feature extracting.

The main disadvantage of image based classification is disturbance from complicated background. For example, in sports competitions scene, components stand too close as some parts of human body may be covered by each other. Besides, the classification model may mix human bodies in background and the target body. In contrast of image based method, the skeleton based method make classification by skeleton data of the target body which could get rid of the influence of complicated background and unrelated bodies.

The main contributions of our frameworks lie in three folds:

- A skeleton-based dataset of boxing technical action is proposed for fine-grained classification in sports area.
- An interpolated method based on Neville interpolation is proposed for the augmentation of low-frame-rate data.
- An improved BoxNet is proposed for better feature fusion of both joint data and bone data in fine-grained technical action classification.

II. RELATED WORK

A. Skeleton-based neural network in general action classification

The skeleton-based method has already been used in widespread areas to handle relationships between nodes which including keypoints data of human body [3]–[6]. Amongst which, graph neural networks has proved to have the best performance. *Yan et al.* firstly introduced graph convolution network to representation study of human action in ST-GCN [7]. In 2s-AGCN [8] by *Shi et al.*, parameterized dynamic link between any nodes in graph was proposed to find more relation between different parts in graph. Contrary, *Thakkar et al.* introduced a kind of pb-GCN [9] which separates entire graph into multiple parts to extract the feature of each body

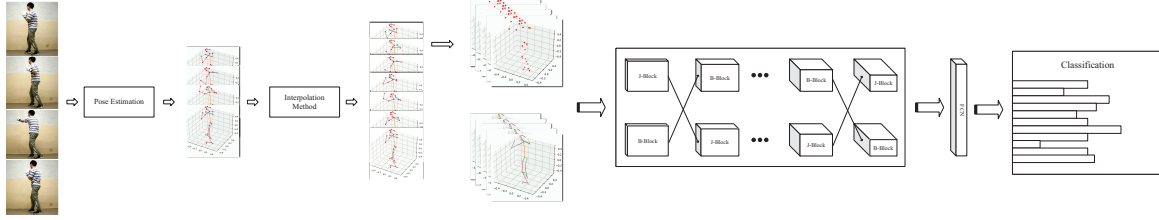


Fig. 1. The pipeline of BoxNet, which has Three steps. (1) Data transformation.(2) Data interpolation. (3) Feature extraction and infusion. (4) Classification. The technical action classifier is the stack of staggered J-TCN and B-TCN blocks.

part independently. Another non-negligible neural network is DGNN [10]. Proposed by *Shi et al.*, DGNN adopted directed acyclic graph which forces the information of each node to transmit from inner to outer part of human body. Meanwhile, it also merge the joints data and bones data in a single stream to extract the feature more effectively and efficiently. However, as DGNN uses the full connection layers in forward propagation, the inference speed of DGNN is inferior to those use convolutional layers.

B. Fine-grained classification

There have been multiple fine-grained method in diverse area such as image classification. In very conventional method, more tiny part locations should be labeled in order to train the network to acquire distinguished feature in these area. To avoid the complex labeling work, *Lin et al.* proposed a bilinear CNN model [11] which consists of two feature extractors whose outputs are multiplied using outer product at each location of the image and pooled to obtain an image descriptor. To simulate the mechanism of human observation, *Ren et al.* introduced vision attention model to image classification. An attention proposal network is applied in RA-CNN [12], which could effectively locate the distinguished region of image.

III. METHOD

In this section, we will introduce the structure and component of our proposed BoxNet in detail.

A. Joint and bone information

Previous work have shown the increasing performance for the combination of joints' and bones' information. Bones' graph of human body can be represented as the difference of nodes' coordinate. For example, the bone which connected with two joints $v_1 = (x_1, y_1, z_1)$ and $v_2 = (x_2, y_2, z_2)$ can be represented as $e = (x_1 - x_2, y_1 - y_2, z_1 - z_2)$.

B. Graph construction

Previous graph convolution network and directed graph neural network both have disadvantage in global feature extracting. These graph convolution networks transmit information of each nodes and edges separately, which means the impossibility for nodes adopting features of edges in model's backbone and vice versa. Contrary, directed neural network

transmit information of nodes and edges from a certain starting node which blocks the interaction between nodes and edges from different sides of starting node. The combination of advantage of these two network allow the information of each node and edge to transmit amongst every node and edge.

For each vertex v_i in human body graph, we define e_i^{inter} as the edge linked to v_i which is closer to the center of human body and e_i^{outer} as the edge which is farther to the center. Similarly, for the edge e_j , we define v_j^{inter} as the vertex which is closer to the center and v_j^{outer} as the vertex which is farther to the center. The set of joints could be represent as $V = \{v_1, v_2, \dots, v_n\}$ and edges could be represented as $E = \{e_1, e_2, \dots, e_m\}$, where n and m denotes the number of joints and edges, separately. And the full sequence is $S = \{G_1, G_2, \dots, G_T\}$, where $G = \{V, E\}$ and T is the length of the sequence.

C. Structure of BoxNet

Since the data is organized in graph form, the problem is how to extract information from the graph for action classification, as we use the dependencies between joints and bones. The solution in previous work is the directed neural network which updates information of each joint together with its adjacency joints by full connection layers, which is shown in figure IV. However, when it comes to the convolution layers, for example, the updated joints itself and its neighbours ought to be operated by different kernels. The diversity of kernels could extract the distinguished feature of the centre part, the updated joints, and its neighbours, bones connect to it.

In sports technical action classification, as the movements of fist which would be hardly affected by that of feet but directly controlled by the wrist, the motion in nearby joints would have more influence to the target joint. Thus, the local feature fusion in neighbourhood ought to have higher priority than that of farther ones. For restrain the joints' information locomotion in the graph, as the movements of feet and hardly influence movements of hands, the depth of the network ought to be limited, which function similarly with directed graph based method, while our method can also make it possible for joints and edges to exchange information within certain extent.

The Temporal convolution network is the basic part for BoxNet. Compared with the normal GCN network block, it

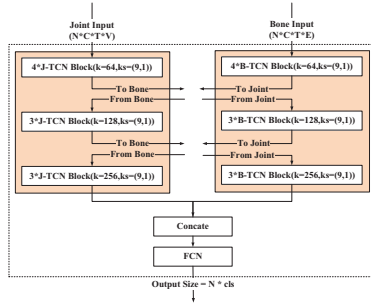


Fig. 2. The detail of J-TCN and B-TCN blocks. Give input bone data and joint data which share same size, the output of each convolution operation would be transmit to the input of next convolution operation would be acquired from the other stream.

has two types of operation, joints updating and edges updating. The function of normal GCN kernel is shown as:

$$f_{out}(v_i) = \sum_{k=0}^K W_k f_{in}(v_i) L \quad (1)$$

$$L = D^{-1/2} (D - A) D^{-1/2} \quad (2)$$

where W_k are different types of kernel, L is the Laplace matrix of the graph, D and A are the degree matrix and the adjacency matrix of the graph separately. The elements in $N_v \times N_v$ adjacency matrix are:

$$A[i][j] = \begin{cases} 1 & v_i \text{ linked to } v_j \\ 0 & v_i \text{ not linked to } v_j \end{cases} \quad (3)$$

where N_v is the number of joints in the graph.

However, when it comes to the information exchange between joints and edges, the updating function should be modified as such:

$$\begin{cases} f_{out}(v_i) = \sum_{k=0}^K W_k f_{in}(e_j) L^{-1} \\ f_{out}(e_i) = \sum_{k=0}^K W_k f_{in}(v_j) L \end{cases} \quad (4)$$

while the adjacency matrix is no more a description of the relation between connected joints but a structure of edges and joints, the $N_v \times N_e$ adjacency matrix ought to be formulated by:

$$A[i][j] = \begin{cases} 1 & v_i \text{ linked to } e_j \\ 0 & v_i \text{ not linked to } e_j \end{cases} \quad (5)$$

Network Architecture. The depth of network is related to receptive field. Take the CNN as an example, the 3×3 kernel in network whose depth is 10 with of pooling operation provides each pixel with receptive field of 11×11 . Similarly, the receptive of field is decided by the depth of network either, the each graphic convolution operation add information of one more node or edge to the operated object. According to the topology of graph, the depth of network should be varied, as the receptive depth ought to match the level of specification

of graph. If there is two other joints between fingertip and shoulder, the depth of network which could provide the joint of fingertip with the information of shoulder should be at least four in our framework.

D. Importance matrix for BoxNet

The influence for each joints and edge to its neighbour varied, which makes it necessary to used a kind of matrix to express the importance of them during information updating procedure. Meanwhile, the manually constructed graph can hardly express the deeper semantic of structure of human body, which means the additional relationship of edges and joints shall be established during information updating. Take the blocking technical action as an example, the naturally unconnected hands would formulate an extra link when blocking the punch of the opposite in boxing. Both *Yan et al.* and *Shi et al.* adopt a kind of attention map to simulate this phenomenon. While A is the original adjacency matrix, we denote IM as the importance map of A . Combined with the updating connection between edges and bones, the non-normalized Laplace matrix $P = A + IM$ could be directly parameterized. What should be pay more attention is that the transmission of joints to edges and edges to joints ought to be considered separately by introducing P_v and P_e . What's more, the parameterized matrix is strongly correlated to the early learning result, which makes it necessary to be fixed when learning process just started.

E. Skeleton temporal augmentation

According to introduction part, we noted that low-frame-rate data collection with high-velocity movements may hind the accurate feature extracting. Skeleton data in a clip is not always be adequate for recognition task, so it is necessary to elevate the frame rate of origin video. As it is widely used in solve data missing problems in aerospace science such as locating satellite orbital position from dispersed data [13], spatial interpolation method may be able to have a satisfying performance in the inserting frame of skeleton-based data. The order of interpolation function represents its predictive power for the tendency of movements. One function which is widely used in above-mentioned area is the Neville interpolation [14], [15] as follow:

$$T_{i,j}(t) = \frac{(t - t_i)T_{i-1,j-1} - (t - t_{i-j+1})T_{i,j-1}}{t_{i-j+1} - t_i} \quad (6)$$

where $T_{i,j}$ are from the difference quotient which i denotes each reference point and j denotes the order of difference quotient. Take $T_{3,2}$ as the example, it is the result of $T_{3,1}$ and $T_{2,1}$, where t_3 and t_2 also involves in this process.

A primitive point which should be paid attention to is the Runge phenomenon which happens near the ends of data clip. It requires longer input data for the integrity of actions, as both ends of enlarged data clip ought to be deleted to avoid the influence of Runge phenomenon.

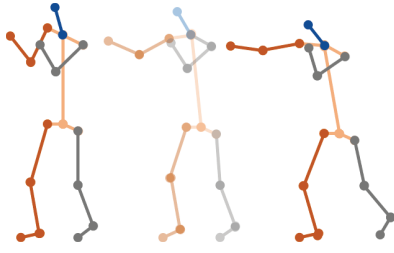


Fig. 3. The schematic of **interpolated Jab action**, as solid skeletons were mapped from RGB video, while transparent one is the interpolated result of previous and following clips

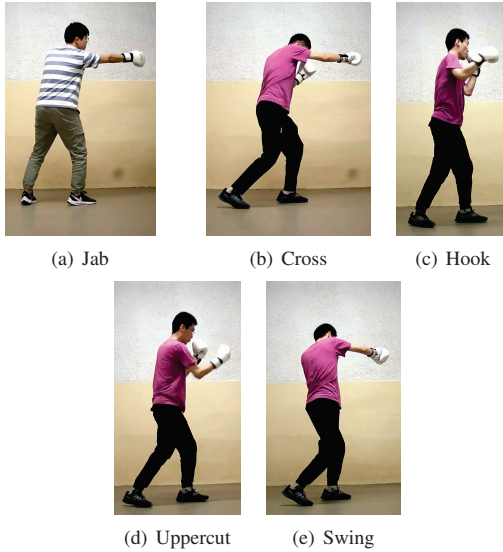


Fig. 4. Examples of Key frames of 5 types of boxing technical action, performed by some members of our team.

IV. BOXING DATASET

BoxNet is trained on boxing dataset, which is collected in laboratory environment. Three members of our team performed 5 types of boxing technical actions for 3 to 5 repetitions each in eight directions, for a total of 1274 videos. As the division of boxing technical actions varied, we chose one popular classification method which sorts atomic boxing action into 5 types as follow [16], [17]:

- **Jab(SJ)**: the lead fist is thrown straight ahead and the arm is fully extended from the side of the torso. Fist is generally held in a horizontal orientation with the palm facing the ground.
- **Cross(SI)**: the rear hand is thrown from the chin side, travelling towards the target in a straight line, the torso and hips rotate counterclockwise as the straight/cross is thrown.
- **Hook(FH)**: the rear fist makes an upward motion that resembles a pirate's hook in shape
- **Uppercut(UC)**: by turning the core muscles and back, thereby swinging the arm, which is bent at an angle near or at 90 degrees, in a horizontal arc into the opponent.

- **Swing(SW)**: the swing the arm is more extended than hook, the palm faces outwardly.

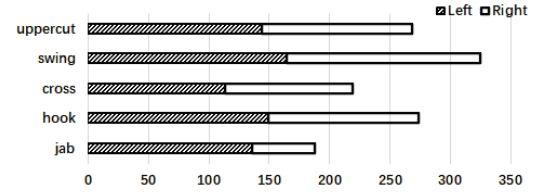


Fig. 5. The number of samples from different classes

TABLE I

THE STATISTICAL DATA OF BOXING DATASET, INCLUDING THE SAMPLE SIZE OF EACH TECHNICAL ACTION AND THE AVERAGE DURATION OF THESE. IT IS NOTABLE THAT ALL KINDS OF TECHNICAL ACTION IS SORTED INTO LEFT FIST RIGHT FIST.

Technical action	Sample size	Average duration(frames)
Jab	188	27.52
Cross	219	33.70
Uppercut	268	40.88
Hook	274	44.39
Swing	325	35.17

Boxing dataset contains RGB data acquired by Hikvision surveillance camera at 30Hz. For this study, we use pose estimation network to obtain necessary 3D skeleton sequences.

V. EXPERIMENT

In this section we evaluate the performance of BoxNet in skeleton-based action recognition experiments, using general dataset and fine-grained dataset either. We experiment on two kinds of dataset, general dataset NTU-RGB+D with a large-scale daily indoor actions and our boxing dataset with 14 kinds of fine-grained boxing technical action. Through experiments on both dataset, we would prove comparable efficiency and effectiveness of proposed framework in general scenario, and the notable performance on fine-grained sports technical action classification. In particular, We first performed detailed ablation study on Boxing dataset dataset to examine the contributions of proposed model components to the classification performance. To verify whether the experience we gained on in the unconstrained setting is universal, we experiment with the constraint setting on Boxing dataset and compare Boxing dataset with other state-of-the-art approaches. All experiments were conducted on PyTorch deep learning framework [18] with 1 RTX-3090 GPU.

A. Data preparation and augmentation

NTU-RGB+D [19] is a large-scale dataset for RGB+D human action recognition with more than 56 thousand video samples and 4 million frames, collected from 40 distinct subjects. It contains 60 different action classes including daily, mutual, and health-related actions. NTU-RGB+D has already been a recognized platform to verify the validity of the model for action classification.

Although NTU-RGB+D provides 3d keypoints data of human action, our boxing dataset was recorded by common surveillance camera, which only provide RGB video with resolution of 1080p and frame rate of 25. So our work use a pose estimation network to generate keypoints data.

VIBE is a prominent work in human body pose and shape estimation, which could generate SMPL model of human body through video inference. Generated SMPL model describe human body by two kinds of parameters, shape parameters and pose parameters. Pose parameters is an 24×3 array which describe 24 joints of human body in 3d spatial domain. Beside, VIBE also provide 2d keypoints estimation results.

As the frame rate of origin data is not high enough to capture information of every moment. Due to the simplicity of skeleton data compared with RGB image, we use interpolation method to add intermediate frame. According to empirical conclusion in aerospace area, we implement 11 order Neville method and delete starting and ending frames to avoid Runge phenomenon.

B. Comparison with common action classification frameworks

We first compared the effects of boxnet with the generic graph convolutional action classification method [7], [8], [10] on the NTU-RGB+D dataset, as shown in Table. We can see

TABLE II
COMPARISON OF THE RESULT OF BOXNET WITH AND WITHOUT DATA AUGMENTATION.

	NTU-RGB+D 60		Box Dataset	
	xsub	xview	Top 1	Top 5
ST-GCN [7]	81.5%	88.3%	65.83%	98%
2S-AGCN [8]	88.5%	95.1%	89.52%	100%
DGNN [10]	89.9%	96.1%	78.75%	99.17%
Ours	87.6%	94.7%	91.25%	100%

that boxnet performs close to SOTA on the generic action dataset and outperforms mainstream graph convolutional networks such as ST-GCN.

Then, we tested boxnet as well as the generic graph convolutional action classification method on BD, and the results are shown in Table. It can be seen that boxnet outperforms the mainstream graph convolutional networks on the fine-grained dataset Box Dataset and can effectively classify fine-grained actions in boxing.

C. Ablation study

We examine the effectiveness of proposed components in BoxNet in this section on Boxing dataset.

1) *Dimension of input data*: Both 3d and 2d keypoints data can be generalized by VIBE, where 3d keypoints data is elevated from 2d data. In this work, we explore the influence of pose estimation network to fine-grained classification result. The original dataset is captured by a monocular surveillance camera. Both 2d and 3d pose estimation result is generated by VIBE, which share same time span and frame rate.

We process generated 2d and 3d data by basical BoxNet, the result is shown in Table IV. The classification result of 2d input

TABLE III
COMPARISON OF THE RESULT OF BOXNET WITH AND WITHOUT DATA AUGMENTATION.

	Top 1	Top 5
inward graph	75.00%	99.58%
outward graph	78.75%	99.17%
bi-direction graph	82.08%	100%
BoxNet without IM	72.00%	99.58%
BoxNet with IM	82.08%	99.13%
with augmentation	91.25%	99.82%

data is a total catastrophe. The network are not able to tell any difference between different boxing technical action. The result shows that the network classified all technical action into one class. Compared with 2d input data, 3d data are able to express the feature of boxing technical action in an acceptable level.

TABLE IV
COMPARISON OF THE RESULT OF 2D AND 3D KEYPOINTS INPUT OF BOXNET

Keypoints	Top 1	Top 5
2d	14.17%	57.92%
3d	76.25%	99.58%

2) *Importance matrix*: Previous work has shown that parameterized adjacency matrix has positive effect on action classification. However, whether it still functions well in fine-grained action classification is unconfirmed. We compared BoxNet with and without parameterized adjacency matrix on multiple types of Boxing dataset and the result is shown in Table III(row 4 and row 5).

3) *Data augmentation*: Proposed interpolation method has ability to add unlimited frame in skeleton data, and we use the Neville method to interpolate the action sequence to three times its original length. The result is shown in Table III(row 6).

4) *Data transmitting strategy*: There are 3 types of data transmitting strategy, i.e., inward, outward and bi-direction. To investigate the performance of these methods, we designed directed graph to limit the transmitting direction of nodes' information. We test our network with different data transmitting strategy on boxing dataset and result is shown in Table III(row 1,2). The outstanding performance of bi-direction strategy provided that human body in technical action has no certain centre point and the information of each node is necessary to others.

5) *Early fusion method*: There are two types of data fusion in GCN, i.e., the early fusion method as DGNN and ours, and two-stream method as ST-GCN and 2s-AGCN. We applied both early fusion and late fusion on our network and the result is shown in Table V.

VI. CONCLUSION

In this paper, we present a dataset of boxing for fine-grained action classification task and an improved GCN network for

TABLE V
COMPARISON OF THE RESULT OF BOXNET WITH AND WITHOUT DATA
AUGMENTATION.

	top 1	top 5
two-stream	70.83%	99.17%
early fusion	82.08%	99.82%

it. The model adopt early fusion of features of bones and joints in GCN. The interpolation method is also introduced to improve the performance of network on low-quality dataset. The BoxNet is firstly implemented on boxing technical action classification and has outstanding performance. A fine-grained dataset of boxing technical action is also established for further study. Further works could focus on the fusion of body-parts' RGB information and graph feature to detect the rotation of joints, which may has vital influence on more detailed sports technical action.

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